

SE VIBECODING LAB

PROMTS:

Name: Annanya Sharma

SRN: PES1UG24CS069

Llm conversation link:

<https://chatgpt.com/share/6ac62982-99f4-83e8-8163-2298b7bc08de>

Task 1 prompt:

I am trying to make a coin collector game using pygame library, inside I want to edit this such that the moment tha player touches the coin, the coin vanishes and does not award more points to the player.

currently I have this function:

```
def update(self):  
    collected = check_collection(self.player, self.coins)  
    for coin in collected:  
        self.score += coin.value
```

this calls this functions:

```
def check_collection(player, coins):  
    player_rect = player.get_rect()  
    collected = []  
    for coin in coins:  
        if player_rect.colliderect(coin.get_rect()):  
            collected.append(coin)  
    return collected
```

how do I edit update() so the coin vanishes after awarding the points only once

TASK 2 Prompt:

currently in my game_engine.py file, I have intialized a global variable COIN_VALUE =1 but I want to change the below functions which are defined in class GameEngine: def **init**(self): self.player = Player(x=WIDTH / 2, y=HEIGHT / 2) self.coins = [self._random_coin() for _ in range(NUM_COINS)] self.score = 0

```
def _random_coin(self):  
    x = random.randint(30, WIDTH - 30)  
    y = random.randint(30, HEIGHT - 30)  
    return Coin(x=x, y=y, radius=12, value=COIN_VALUE)
```

such that there are 3 coin types with different color and value (bronze-> 1 point, silver-> 3 points and gold -> 5 points). Whenever the player collides with a coin, the player should get the coin's respective points added.

TASK 3 prompt:

Now I want to add obstacles to the game,

1. I want to make a new file where I define a class for obstacles which are red and triangular. and it should have its own respective hitbox the same way coin has one. the coin code is: class Coin: def **init**(self, x, y,value, color,radius=12): self.x = x self.y = y self.radius = radius self.value = value self.color = color def get_rect(self): return pygame.Rect(int(self.x - self.radius), int(self.y - self.radius), self.radius * 2, self.radius * 2,)
2. I want the obstacles to be rendered in the renderer.py file function: def draw_scene(surface, player, coins): surface.fill(COLOR_BG) for coin in coins: pygame.draw.circle(surface, coin.color, (int(coin.x), int(coin.y)), coin.radius) pygame.draw.rect(surface, COLOR_PLAYER, player.get_rect(), border_radius=4)
3. in game_engine, I want to edit these two files such that the obstacles are shown and if the player collides with any obstacle, it removes all coins, obstacles and player from the screen. def update(self): collected = check_collection(self.player, self.coins) for coin in collected: self.score += coin.value self.coins.remove(coin) def

```
draw(self, surface, font): from game import renderer renderer.draw_scene(surface,
self.player, self.coins) renderer.draw_text(surface, font, f"Score: {self.score}", (10,
10))
```

4. in renderer file again, I want to edit this function so that it shows "GAME OVER" and the score when the player hits an obstacle
- ```
def draw_banner(surface, font, text):
 surf = font.render(text, True, (255, 220, 80))
 rect = surf.get_rect(center=(surface.get_width() // 2, surface.get_height() // 2))
 surface.blit(surf, rect)
```

#### **TASK 4 Prompt:**

I want to now:

- 1) change the obstacle shape to a rectangle instead so it matches with its hitbox
- 2) Add 3 player lives, show them beside score as : Lives left: 2 or Lives left: 3, which decreases by 1 each time the player hits an obstacle and then gives the player 5 seconds to move away from obstacle before it considers the collision as another hit.
- 3) show a countdown (30 second countdown) beside score which decrements by 1 each second
- 3) change the code such that once the lives left =1 and the layer hits the obstacle OR the countdown reaches 0, then all the sprites disappear, the score is shown, "GAME OVER" shows up in bold fonts on the screen and also a way to restart the game